

Valuation and Compensation of Energy Not Supplied

Leonardo Gabriel Granda Erazo 

Universidad Nacional de Loja, Loja, Ecuador

leonardo.granda@unl.edu.ec

Abstract—The growing dependence of modern societies on electricity requires reliable estimates of the Cost of Energy Not Supplied (CENS) to support infrastructure investment and establish effective service-quality regulation. This article examines CENS through a comparative analysis of the United States, the United Kingdom, Chile, Peru, and Ecuador, with emphasis on their valuation methods, regulatory frameworks, and compensation mechanisms. The analysis considers two principal criteria: economic efficiency and regulatory simplicity. Although estimated values differ substantially, all of the systems examined rely on quality indicators to establish compensation or penalty thresholds. Practical cases further demonstrate that the high cost of prolonged-interruption compensation provides a strong incentive for distribution companies to prioritize preventive network investment over the payment of penalties to customers.

Index Terms—Energy Not Supplied (ENS), Cost of Energy Not Supplied (CENS), electricity regulation, supply quality, customer compensation, infrastructure investment.

I. INTRODUCTION

THE reliability and quality of the electricity supply are essential pillars of modern economic development. According to the literature [1], [2], electricity-supply quality can be divided into three principal dimensions: availability (service continuity), technical characteristics (voltage quality), and commercial quality (management of customer requests).

In this context, Energy Not Supplied (ENS) is a critical indicator that quantifies the electricity not delivered to users because of infrastructure failures, scheduled interruptions, or insufficient generation and distribution capacity. It therefore provides a quantitative representation of economic losses and may be expressed in USD/interruption, USD/kW of peak load lost, USD/kWh of annual energy consumed, or USD/kWh of energy not supplied [3].

ENS concerns service continuity, whose central objective is the uninterrupted provision of electricity. To assess network performance, international standards such as IEEE 1366 [4] define widely recognized indicators, including:

- SAIDI (System Average Interruption Duration Index): the average total duration of interruptions experienced by a customer during a specified period.

$$\text{SAIDI} = \frac{\sum_{i=1}^n N_i t_i}{N_T} \quad (1)$$

- SAIFI (System Average Interruption Frequency Index): the average frequency of interruptions experienced by a customer during a specified period.

$$\text{SAIFI} = \frac{\sum_{i=1}^n N_i}{N_T} \quad (2)$$

Other indicators include the Customer Average Interruption Frequency Index (CAIFI) and the Customer Average Interruption Duration Index (CAIDI) [4], [5]. The literature also employs measures such as IEAR, VoLL, and WTP [3]; nevertheless, SAIDI and SAIFI remain the most widely used.

When these indicators exceed acceptable thresholds, the consequences are borne directly by users and the wider economy. As noted in [6], outages and voltage fluctuations in developing countries cause material losses and increase production costs. The associated economic effects may be classified as direct—including lost production, process-related costs, and equipment damage—or indirect, such as deferred income and reduced household comfort.

To determine the optimal level of quality, society must balance the cost of providing that quality through network investment against the interruption-related costs borne by customers [2].

Given the social and economic cost of ENS, accurate estimates not only inform the planning of new infrastructure but also underpin consumer-compensation policies intended to balance operator accountability with user protection.

Compensation for prolonged interruptions, whether provided through bill credits or fixed payments, is intended to create a direct incentive for distribution companies to improve network quality. Its implementation, however, differs considerably among countries. Some jurisdictions employ methodologies based on aggregate indices, whereas others have adopted individualized systems that calculate ENS for each customer.

This diversity has generated regulatory debate concerning equity and efficiency. Approaches range from direct and transparent payment models to regulated and often indirect systems in South America.

This article reviews methods for estimating the Cost of Energy Not Supplied (CENS) and the regulatory mechanisms used to compensate customers. Two groups of cases are considered: South American and international cases.

The remainder of this paper first examines CENS estimation and its calculation methods, including probabilistic models such as EENS and exclusions for force-majeure events. It then discusses valuation approaches in international and South American contexts, followed by practical compensation cases. The Discussion compares the rigor of the regulatory frameworks and draws conclusions regarding the role of ENS in the resilience of the electricity sector.

II. ESTIMATION OF THE COST OF ENERGY NOT SUPPLIED

Estimating the Cost of Energy Not Supplied (CENS) entails quantifying the economic and social losses associated with

interruptions. The corresponding value is calculated as the product of interrupted power and interruption duration, with adjustments to exclude scheduled events and events classified as force majeure.

According to [7], the value that distribution companies assign to supply reliability is a key factor in efficient decision-making. Because this value depends on unobservable variables, its calculation requires methodologies based on approximate estimates.

Valuation methodologies vary according to their purpose—academic or regulatory—and the maturity of the market. Countries with advanced monitoring systems, such as the United Kingdom and the United States, can produce more precise planning estimates. In South America, by contrast, regulatory compensation is generally determined through metrics that directly incorporate customer-impact indices [6].

Two considerations govern the desired level of confidence in service continuity. Greater reliability generally requires substantial infrastructure investment, whereas reduced reliability may require economic compensation to users because interruptions become more frequent. Consequently, insufficient investment may produce unreliable supply, while excessive investment may result in unnecessary expenditure [3].

For these reasons, [7] argues that the value assigned to electricity-supply continuity must be estimated in order to minimize the total net cost of reliability, as illustrated in Fig. 1.

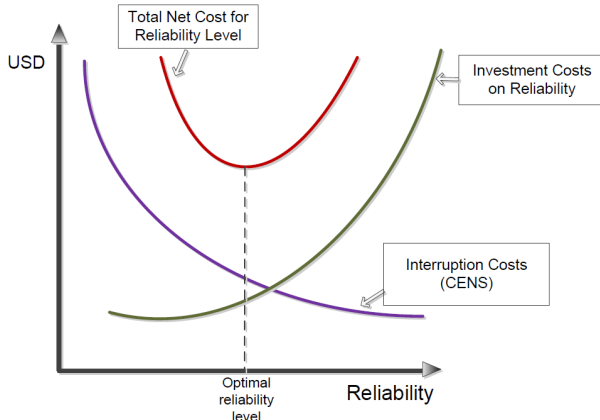


Fig. 1. Total net cost of the reliability level. Adapted from [7].

More specifically, CENS can be organized around two broad conceptual models: the economic approach and the deterministic regulatory approach.

The economic approach, also referred to as service-cost valuation, seeks primarily to model the actual economic loss experienced by users. Interruption-cost studies under this approach rely on broad sample surveys (survey-based methods) and detailed analyses of representative situations (case-based methods) [2], [3].

The resulting survey data are used to construct Customer Damage Functions (CDFs) [3], [7], [8]. These functions enable a detailed valuation of ENS by disaggregating costs according to variables such as consumption sector (residential, commercial, or industrial), event duration, season, time of day, and customer characteristics [8].

Reference [2] presents and classifies several cost-estimation methods, of which the following are particularly relevant:

- Bottom-up methods: detailed cost data are collected at the user level and subsequently aggregated to obtain sectoral or overall estimates.
- Top-down methods: available macroeconomic data, such as sectoral GDP losses, are used to produce aggregate approximations.
- Direct methods: surveys or market studies explicitly assess the costs incurred by customers or their willingness to pay or accept compensation.
- Indirect methods: customer preferences and priorities are inferred without directly eliciting a monetary value for the quality problem.

Although surveys are fundamental valuation tools, they are rarely used directly to impose regulatory penalties. Respondents may negatively assess the service if they perceive that doing so could produce a direct benefit, such as a penalty against the utility, thereby compromising the objectivity of the valuation [6].

In countries such as the United States, where survey methods are common, heterogeneous sectoral data must be normalized using factors such as annual consumption, average load, or energy not supplied [2]. The individual normalized data are then combined to construct the customer damage function:

- Individual normalized cost:

$$c_{N,i}(r, t) = \frac{C_i(r, t)}{N_i(r, t)} \quad (3)$$

where C_i is the monetary value of the interruption for respondent i (in currency units), and N_i is the normalization factor (in kWh or kW).

- Sector Customer Damage Function:

$$c_{SCDF}(r, t) = \frac{1}{m} \sum_{i=1}^m c_{N,i}(r, t) \quad (4)$$

c_{SCDF} is calculated as the mean normalized cost for the m respondents in sector s , yielding a cost per unit of energy or power (currency/kWh or currency/kW).

These expressions are proposed in [2]. Although precise estimation is complex and varies among countries, it is essential for identifying the optimal regulatory balance in supply quality.

In contrast to economic service-valuation models, most regulatory frameworks use deterministic methodologies. Under this approach, the obligation to compensate arises when predefined quality limits are exceeded; regulatory oversight and enforceable penalties are prioritized over determining the exact economic cost of an interruption for each customer [8].

According to the review in [9], before Chile established the National Electricity System (SEN), its power system was divided into the Central Interconnected System (SIC) and the Northern Interconnected System (SING). The study estimated unit ENS costs of 3.24 USD/kWh and 4.86 USD/kWh for the two systems, respectively, while noting that these values were subject to revision and updating.

The current Chilean model for calculating unauthorized interrupted energy is deterministic and directly relates the customer's average demand to the interruption duration. It is activated when the applicable duration and frequency limits are exceeded [10]. For regulated customers, the expression is:

$$E_{int_{i,r,s,e}} = F_{i,r,s,e} \sum_{m=1}^n cph_m \cdot TTINA_{m,r,s,e} \quad (5)$$

where F is the interruption-time allocation factor, cph_m is the average hourly demand of customer m , and $TTINA$ is the unauthorized interruption time in hours. If both duration and frequency limits are exceeded during the same event, the index that produces the greater compensation is applied.

Ecuador also illustrates a deterministic system, because transmission and connection quality are evaluated through interruption duration and frequency [9]. The Ministry of Environment and Energy establishes a CENS value of 1533 USD/MWh; however, this figure is used exclusively to quantify risk and justify infrastructure investment, rather than to compensate customers [11].

Although Ecuador does not have a methodology for determining CENS, it defines expressions for ENS. The mechanism is deterministic because it is activated only when the service-quality indices FMIK and TTIK exceed their predefined limits. CONELEC Regulation [12] establishes four formulas according to whether the frequency and duration limits are exceeded:

- If $FMIK > LimFMIK$ and $TTIK < LimTTIK$:

$$ENS = (FMIK - LimFMIK) \cdot \frac{TTIK}{FMIK} \cdot \frac{ETF}{THPA} \quad (6)$$

- If $FMIK < LimFMIK$ and $TTIK > LimTTIK$:

$$ENS = (TTIK - LimTTIK) \cdot \frac{ETF}{THPA} \quad (7)$$

- If $FMIK > LimFMIK$ and $TTIK > LimTTIK$, and if $\frac{TTIK}{FMIK} < \frac{LimTTIK}{LimFMIK}$:

$$ENS = (FMIK - LimFMIK) \cdot \frac{TTIK}{FMIK} \cdot \frac{ETF}{THPA} \quad (8)$$

- If $FMIK > LimFMIK$ and $TTIK > LimTTIK$, and if $\frac{TTIK}{FMIK} \geq \frac{LimTTIK}{LimFMIK}$:

$$ENS = (TTIK - LimTTIK) \cdot \frac{ETF}{THPA} \quad (9)$$

where ETF is the total billed energy and $THPA$ is the duration, in hours, of the analysis period.

III. VALUATION APPROACHES

The performance of electricity systems varies significantly worldwide, resulting in different approaches to the valuation and regulation of CENS.

In terms of continuity, Latin America exhibits higher quality levels than other developing regions. A 2016 study reports that firms in Latin America experience approximately three interruptions per month, substantially fewer than the approximately 25 monthly interruptions reported in South Asia [6]. However, the same study indicates that interruptions in Latin America tend to last longer than those recorded in Europe and Central Asia, as shown in Fig. 2.

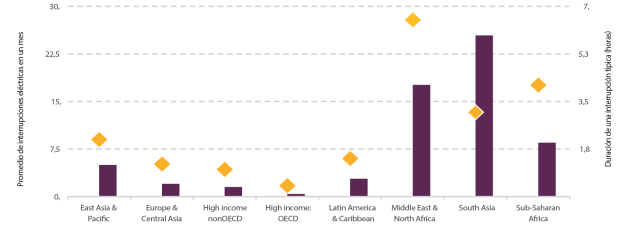


Fig. 2. Duration and frequency of interruptions worldwide. Adapted from [6].

Despite this relative advantage, service-quality patterns and the value of interruption-related losses are not homogeneous within Latin America. Outage frequency and the associated CENS vary considerably among countries, thereby supporting the need for regulatory frameworks tailored to national circumstances, as illustrated in Fig. 3.

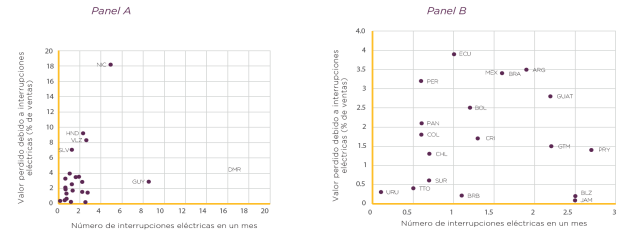


Fig. 3. Interruptions and their associated value of lost service. Adapted from [6].

To ensure desirable levels of electricity-service performance, regulators generally employ four principal instruments [6]:

- publication of company-performance data;
- establishment of minimum quality standards;
- implementation of reward and penalty mechanisms; and
- promotion of higher-quality contracts between utilities and their customers.

Because of their distinct histories and institutional development, countries have implemented regulations based on different methodologies and definitions of quality. This lack of uniformity complicates the implementation, monitoring, and enforcement of quality indicators and frequently prevents direct international comparison.

The following subsections examine the valuation approaches adopted by selected countries globally and in South America.

A. United States (US)

The United States electricity system addresses interruption risk and the valuation of unserved supply primarily through the Value of Lost Load (VoLL), which is equivalent to the Cost of Energy Not Supplied (CENS). VoLL is a key parameter in system planning and energy-market price caps and is expressed in dollars per kilowatt-hour or dollars per megawatt-hour not supplied.

Supply-reliability regulation combines federal and state-level responsibilities:

- Federal level: the Federal Energy Regulatory Commission (FERC) regulates wholesale markets, while the North

American Electric Reliability Corporation (NERC) establishes mandatory reliability standards. These standards define general planning criteria but do not directly set a price for unserved energy. The Department of Energy (DOE) provides technical analysis and has proposed very low ENS thresholds (e.g., normalized unserved energy less than or equal to 0.002%) [13], [14].

- State level: state Public Utility Commissions (PUCs) oversee customer service and establish their own reliability objectives. Many states use a Loss of Load Expectation (LOLE) criterion of 0.1, equivalent to one loss-of-load event every ten years [14].

In summary, no single federal law requires compensation for ENS. Instead, the system relies on probabilistic reliability standards and market mechanisms, including price caps, established under federal and state regulatory arrangements.

VoLL is the principal valuation measure and is commonly derived from willingness-to-pay studies. It is dynamic rather than fixed nationally and varies significantly according to the circumstances and customer category, thereby requiring disaggregated analysis.

As in other countries, VoLL is adjusted according to customer type, outage duration, interruption frequency, time of day, availability of backup supply, and advance notice [15], [16]. In general, VoLL increases as outages become longer or more frequent [16].

Studies show that economic losses are substantially greater for commercial and industrial customers, whose productive activities may cease, than for residential customers [15].

Methodologies for monetizing ENS through VoLL generally combine statistical analysis with market studies. They may be grouped into the following categories [15], [17]:

- survey-based methods;
- market-based methods;
- economic and hybrid models, including mathematical and economic analyses and survey-based hybrid models; and
- ex-post impact analyses, which directly estimate costs after actual events.

In addition, organizations such as NERC and LBNL have used customer damage functions that relate outage duration and timing to expected economic losses. This analysis combines probabilistic-assessment data with VoLL studies, demonstrating that valuation is dynamic and seeks to balance risk and cost.

Although the federal system prioritizes prevention because of the substantial annual losses associated with outages, state PUCs may provide direct compensation for prolonged interruptions, often through fixed payments or reimbursement of actual damages. States with compensation programs include Illinois, Michigan, New York, and California [18].

B. United Kingdom (UK)

The United Kingdom's regulatory system, overseen by the Office of Gas and Electricity Markets (Ofgem), operates under the Guaranteed Standards of Performance (GSoPs) [19]. This model prioritizes administrative simplicity and certainty of

compensation rather than a granular economic valuation of ENS for each customer.

Network operators in the United Kingdom are directly responsible for maintaining supply and restoring service after outages. When an interruption occurs, customers are entitled to compensation under the GSoPs framework [20].

Compensation consists of fixed payments based on interruption duration and the classification of the customer or event:

- Standard interruptions: the local network company has between 12 and 24 hours to restore supply, depending on the cause of the failure and the number of properties affected.
 - Domestic customers: if a property remains without electricity for more than 12 hours, the customer is entitled to £95.
 - Non-domestic customers: businesses with non-domestic energy contracts may receive £190 for the initial interruption.
 - Additional compensation: customers receive a further £45 for each additional 12-hour period, up to a maximum of £390.
- Severe-weather compensation: payment levels and qualifying time thresholds vary with event severity, which is divided into two storm categories [20].
 - Category 1 storm: initial compensation is £85 when the outage exceeds 24 hours, followed by £45 for each additional six-hour period, up to a total maximum of £2165.
 - Category 2 storm: initial compensation is £85 when the outage exceeds 48 hours, followed by £45 for each additional six-hour period, up to a total maximum of £2165.

By using standardized fixed amounts, the British model facilitates prompt customer compensation and operates as a threshold-based penalty for the distribution company.

C. Chile (CH)

The Chilean regulatory framework establishes that any event or failure causing an unauthorized loss of supply that exceeds quality standards gives rise to automatic customer compensation, without prejudice to any additional sanctions prescribed by law [10].

The system evaluates quality using SAIDI and SAIFI and applies a deterministic approach to define compensable ENS [6].

Compensable ENS is defined as the energy associated with the period beyond the initial 15-minute threshold that the system coordinator requires to restore service.

The framework nevertheless excludes events considered authorized or outside the distribution company's direct responsibility. The principal exclusions are [10]:

- demand reductions resulting from the provision of ancillary services;
- events caused by unforeseeable circumstances or force majeure;
- interruptions arising from rationing decrees;

- exceptions associated with an energy emergency; and
- scheduled disconnections.

The Chilean compensation model applies a fixed multiplier to the energy tariff. This multiplier is intended to provide a strong investment signal that encourages distribution companies to improve network reliability. The mechanism seeks to enhance service quality by imposing a substantial financial penalty for each unit of unauthorized ENS.

For regulated customers, compensation is equal to the ENS valued at fifteen times the energy tariff in effect during the supply interruption. For unregulated customers, compensation is likewise calculated at fifteen times the energy component of the average market price, obtained by subtracting the capacity component from the basic average price [10].

The regulation establishes an annual ceiling on compensation—generally 5% of the company’s regulated or total revenue in the preceding year—to protect utilities’ financial stability in the event of a catastrophe.

To substantiate the economic value of supply deficits and the severity of the compensation mechanism, the National Energy Commission (CNE) must periodically, generally every four years, conduct short- and long-duration outage-cost studies [21].

These studies examine the behavior of industrial customers and the economic effects of supply deficits, with customers segmented by size, activity, and location. Mathematical tools are also used to represent the effects of electricity deprivation on commercial and residential customers. The resulting costs are updated during each tariff-setting process to reflect changes in their principal components.

D. Peru (PE)

The Peruvian regulatory system, overseen by the Supervisory Agency for Investment in Energy and Mining (Osinergmin), bases the obligation to compensate on noncompliance with the quality standards established in the Technical Standard for the Quality of Electricity Services (NTCSE) and the Electricity Concessions Law [6].

Electricity-supply quality is evaluated in terms of service continuity over a six-month control period [22]. The assessment includes collective indicators of distribution-company performance, such as SAIFI and SAIDI, as well as the following individual indicators:

- Number of interruptions per customer, N . Scheduled interruptions for network expansion or reinforcement are included with a weighting factor of 50%, thereby partially penalizing these events.
- Total Weighted Interruption Duration per customer, D . This indicator is the sum of the individual durations of all interruptions after each duration is adjusted by a weighting factor. A factor of 0.25 is used for expansion-related interruptions and 0.5 for maintenance interruptions.

Under Article 86 of the Electricity Concessions Law, the concessionaire must compensate users when supply is totally or partially interrupted for more than four consecutive hours [22].

The NTCSE requires suppliers to compensate customers when service quality fails to satisfy the prescribed tolerances. Compensation is determined using the following expression [23]:

$$\text{Compensation} = e * E * \text{ENS} \quad (10)$$

where e equals 0.35 US\$/kWh in Stage III, and E is a factor determined by the applicable ranges of N and D .

E. Ecuador (EC)

At the national level, the system operator CENACE monitors ENS and disconnection hours. In 2024, 78,224.8 MWh of ENS was recorded as a result of failures and maintenance, with the distribution shown in Fig. 4. Total disconnection time reached 23.4 hours, an increase of 17 hours relative to the preceding year. These operational metrics include only disconnections of at least 5 MW. The increase was partly attributable to three system-collapse events recorded in the National Interconnected System (SNI) during that year [24].

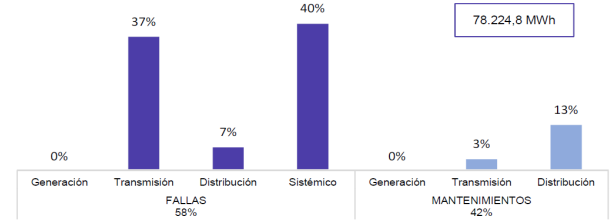


Fig. 4. Energy not supplied during 2024. Adapted from [24].

Unlike Chile and Peru, where the primary mechanism is a direct customer-tariff credit after specified thresholds are exceeded, the Ecuadorian system, supervised by ARCONEL, focuses on penalizing distribution companies for noncompliance with established quality indices.

The electricity-supply quality indices evaluated in Ecuador are FMIK and TTIK [1], [6].

When an electricity company fails to comply with the quality indices, the initial monetary penalty is 20 Unified Basic Salaries (SBU), increasing to 30 SBU for a repeated violation. By using the SBU as the penalty unit, this approach prioritizes sanctions against the company over the direct and automatic customer compensation mechanisms found in other South American countries.

IV. PRACTICAL COMPENSATION CASES

The practical application of compensation models is reflected in the mechanisms employed and the amounts actually paid to customers. The countries examined below exhibit different degrees of automation, public visibility, and compensation coverage.

The United States has no uniform federal ENS-compensation policy, and practices vary substantially among states and utilities. Utilities are generally required to compensate customers under specified conditions, but information concerning such payments receives limited public attention except in major litigation.

Pacific Gas and Electric Company (PG&E) in California provides an example of standardized compensation. Through programs such as the “Safety Net” and the “Storm Inconvenience Bill Credit,” PG&E provides automatic payments to residential customers when an outage caused by a major storm or weather event continues for an extended period [25]. The fixed amounts are intended to mitigate inconvenience rather than cover the full economic loss. Credits range from \$25 to \$100 for interruptions lasting from 48 hours to more than 120 hours and are applied directly to the customer’s bill.

Beyond automatic credits, litigation and class actions are important mechanisms for obtaining substantial financial redress, particularly when utility negligence or systemic failures are established. Following the 2021 Texas energy crisis, for example, class-action claims involving allegedly excessive prices charged by providers such as Griddy Energy reached US\$29.1 million [26].

The British regulatory framework, by contrast, is distinguished by the automation and certainty of compensation under the GSoPs [20]. In addition to outage compensation, Ofgem requires companies to compensate customers for administrative errors. In one case, ten energy suppliers were required to pay more than £7 million to over 34,000 customers because of incorrect technical overcharges [27], [28].

As previously noted, the Chilean system applies a multiplier equal to fifteen times the energy tariff for regulated customers [10]. For large-scale or systemic events, however, the National Consumer Service (Sernac) may initiate Voluntary Collective Procedures (PVCs) to secure compensation on a broad scale.

One example was the agreement reached between Sernac, Enel Distribución, and Enel Colina following the service interruptions of August 2024 [29], [30]. The agreement established collective compensation for 801,141 residential customers.

Payment amounts varied with interruption duration. Customers affected by outages lasting two to six days, for example, received between 36,000 and 40,000 Chilean pesos.

An additional fixed payment of 18,000 Chilean pesos was established for customers who remained without supply for at least 36 consecutive hours and submitted claims for lost food, medicine, or other refrigerated or frozen goods [30].

Although the Peruvian framework supervised by Osinergmin requires electricity companies to compensate users when ENS exceeds established quality limits, publicly reported cases of large-scale automatic compensation are limited. Because these payments generally take the form of automatic bill credits, they receive relatively little public attention.

Audit reports issued by the Office of the Comptroller General nevertheless confirm that these payments are mandatory. One report identified an economic loss incurred by Empresa Regional de Servicio Público de Electricidad del Norte S.A. (Ensa S.A.) after Osinergmin fined the company for failing to compensate its customers. The company was required to compensate 13,755 supply points that exceeded interruption standards in January 2016, for a total of US\$68,057 [31].

In Ecuador, large-scale customer compensation has been implemented as a direct economic-relief measure by the National Government in response to supply crises involving outages of up to 14 hours per day. Rather than applying

the regulated ENS formula for noncompliance with quality indices, the Government adopted a simplified measure with broad coverage: a monthly credit equivalent to 180 kWh, applied directly to residential customers’ electricity bills [32].

The measure was applied for four months—November and December 2024 and January and February 2025. According to ARCONEL calculations, its total cost was approximately US\$174.93 million [33].

The measure was intended to alleviate the direct economic effects of prolonged outages. Customers consuming less than 180 kWh incurred no charge for energy consumption and, in many cases, paid only complementary service charges, such as public lighting and fire-service fees.

V. DISCUSSION

The comparison reveals a fundamental disparity in the treatment of ENS. South American systems are generally governed by quality regulations and indices and employ fixed formulas or deterministic multipliers applied to energy tariffs. This structure facilitates regulatory enforcement and prompt payment but provides a less precise representation of the interruption’s actual economic cost.

The practical cases further show that standardized regulatory mechanisms are often supplemented or replaced by political and collective measures during large-scale crises. Whereas the United Kingdom addresses prolonged interruptions through fixed GSoP payments, Chile uses Sernac Voluntary Collective Procedures and Ecuador has implemented government-sponsored credits equivalent to 180 kWh. These ex-post measures may impose substantial costs on distribution companies or the public sector. Although they lack the precision of a VoLL calculation, they provide a prompt and visible response to losses in social welfare. Restoring trust and mitigating collective harm may therefore take precedence over strict application of the standard regulatory formula.

The considerable financial and reputational costs of mass compensation encourage a preventive, investment-oriented approach. Innovation and infrastructure improvement intended to ensure continuity of supply ex ante are preferable to compensation expenditure and reputational losses ex post.

VI. CONCLUSION

The increasing dependence of modern societies on electricity heightens their vulnerability to interruptions and demonstrates that efficient infrastructure-investment decisions require a clear understanding of the value of energy not supplied. This value monetizes energy security and enables regulators to establish investment limits and scarcity prices. Nevertheless, most CENS estimates are not directly comparable internationally because methodologies, modeling scenarios, and region-specific factors—including customer, industry, and market characteristics—differ substantially. Despite this methodological variation, survey-based approaches can provide a representative estimate of customers’ willingness to pay. Such estimates should be complemented by probabilistic methods and analyses of interruption indices, including SAIFI, SAIDI, and ENS, so that the assessment does not rely solely on aggregate

indices and instead incorporates the variables relevant to a specific calculation.

The effectiveness of service-quality policy is directly related to institutional strength. A regulatory framework must be clear and must provide incentives that enable utilities to sustain the necessary managerial and financial effort. International experience demonstrates a relationship between robust regulation, the institutions required to enforce it, and improvements in the quality of the final service.

Customer compensation, whether determined through fixed ENS multipliers or standardized duration-based payments, constitutes the most visible and costly signal to utilities and encourages them to prioritize preventive infrastructure investment over the payment of penalties.

VII. REFERENCES

- [1] *Regulación nro. arconel 009/2024, Calidad del servicio de distribución y comercialización de energía eléctrica*, Agencia de Regulación y Control de Electricidad (ARCONEL), Quito, Ecuador, 2024. [Online]. Available: <https://arconel.gob.ec/wp-content/uploads/downloads/2024/12/Regulacion-Nro.-ARCONEL-009-24-signed.pdf>
- [2] M. Hofmann, H. Seljeseth, G. H. Volden, and G. Kjølle, "Study on estimation of costs due to electricity interruptions and voltage disturbances," *SINTEF Energy Research*, 2010.
- [3] I. L.-E. SpA and O. B.-E. SrL, "Regulation of continuity of supply in the electricity sector and cost of energy not supplied," *Regulation of continuity of supply in the electricity sector and cost of energy not supplied*, 2009.
- [4] IEEE, "Ieee guide for electric power distribution reliability indices," *IEEE Std 1366-2022 (Revision of IEEE Std 1366-2012)*, pp. 1–44, 2022. DOI: 10.1109/IEEESTD.2022.9955492
- [5] S. D. N. Callejas and J. C. P. Gómez, "A methodology to calculate the non-supplied energy level in distribution grids with radial structure," *Transactions on Energy Systems and Engineering Applications*, vol. 2, no. 2, pp. 23–32, 2021.
- [6] A. Levy and J. J. Carrasco, "Calidad y confiabilidad de los servicios eléctricos en américa latina," *Banco Interamericano de Desarrollo*, 2020.
- [7] P. Vásquez and Á. Vaca, "Methodology for estimating the cost of energy not supplied-ecuadorian case," in *2012 Sixth IEEE/PES Transmission and Distribution: Latin America Conference and Exposition (T&D-LA)*, IEEE, 2012, pp. 1–9.
- [8] M. Sullivan, M. Mercurio, J. Schellenberg, and M. Freeman, "Estimated value of service reliability for electric utility customers in the united states (ernest orlando lawrence berkeley national laboratory, 2009)," *Report LBNL-2132E*, 2016.
- [9] G. Maila and C. Arellano, "Aplicación de la regulación conelec-003/08 "calidad del transporte de electricidad y del servicio de transporte y conexión en el sistema nacional interconectado" y benchmarking de las instalaciones del sistema nacional de transmisión del ecuador," *Revista Técnica "energía"*, vol. 8, no. 1, pp. 68–75, 2012.
- [10] *Norma técnica de indisponibilidad de suministro y compensaciones*, Comisión Nacional de Energía (CNE), Santiago de Chile, 2020. [Online]. Available: <https://www.cne.cl/wp-content/uploads/2021/03/Norma-Tecnica-de-Indisponibilidad-de-Suministro-y-Compensaciones-.pdf>
- [11] Ministerio de Energía y Minas (MEM), *Plan maestro de electricidad 2023-2032*, Informe Técnico, Consultado en la página 129, Sección 4.6.2: Estudios energéticos para la actualización del PEG 2023-2032., Quito, Ecuador, 2023. Accessed: Oct. 20, 2025. [Online]. Available: <https://www.ambienteyenergia.gob.ec/plan-maestro-de-electricidad/>
- [12] *Regulación no. conelec - 004/01, Calidad del servicio eléctrico de distribución*, Aprobada mediante Resolución No. 0116/01, Consejo Nacional de Electricidad (CONELEC), Quito, Ecuador, 2001. [Online]. Available: <https://arconel.gob.ec/wp-content/uploads/downloads/2025/04/Regulacion-No.-CONELEC-004-01.pdf>
- [13] Federal Energy Regulatory Commission (FERC). "Reliability explainer." Página de información sobre la confiabilidad de la red eléctrica en EE. UU., Accessed: Oct. 23, 2025. [Online]. Available: <https://www.ferc.gov/reliability-explainer>
- [14] U.S. Department of Energy (DOE), "Resource adequacy report: Evaluating the reliability and security of the united states electric grid," U.S. Department of Energy, Report, 2025. Accessed: Oct. 25, 2025. [Online]. Available: <https://www.energy.gov/sites/default/files/2025-07/DOE%20Final%20EO%20Report%20%28FINAL%20JULY%207%29.pdf>
- [15] C. Woo, A. Tishler, J. Zarnikau, and Y. Chen, "Average residential outage cost estimates for the lower 48 states in the us," *Energy Economics*, vol. 98, p. 105 270, 2021, ISSN: 0140-9883. DOI: <https://doi.org/10.1016/j.eneco.2021.105270> [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0140988321001754>
- [16] C. Gibbons and S. Sergici, "Value of lost load study for the ercot region," The Brattle Group, Study Final Report, 2024. Accessed: Oct. 25, 2025. [Online]. Available: <https://www.brattle.com/wp-content/uploads/2024/09/Value-of-Lost-Load-Study-for-the-ERCOT-Region.pdf>
- [17] J. P. Carvallo. "The value of lost load: Concepts, methods, and applications." Presentation to Entergy Regional States Committee Working Group, Lawrence Berkeley National Laboratory (LBNL), Accessed: Oct. 25, 2025. [Online]. Available: <https://cdn.misoenergy.org/20240927%20ERSC%20WG%20Item%2003%20The%20Value%20of%20Lost%20Load649514.pdf>
- [18] K. Costello, "Should public utilities compensate customers for service interruptions?" National Regulatory Research Institute (NRRRI), Report, 2012. [Online].

- Available: <https://pubs.naruc.org/pub/FA86AD59-0662-E1F4-1213-3698BF336139>
- [19] National Grid. "Standards of performance." Consultado el 23 de octubre de 2025, Accessed: Oct. 23, 2025. [Online]. Available: <https://customer.nationalgrid.co.uk/advice-and-guidance/standards-of-performance>
- [20] Ofgem (Office of Gas and Electricity Markets). "Check compensation rules for a power cut or supply problem," Accessed: Oct. 23, 2025. [Online]. Available: <https://www.ofgem.gov.uk/information-consumers/energy-advice-households/check-compensation-rules-power-cut-or-supply-problem>
- [21] Comisión Nacional de Energía (CNE). "Costos de falla." Página de información técnica de regulación eléctrica., Gobierno de Chile, Accessed: Oct. 23, 2025. [Online]. Available: <https://www.cne.cl/tarificacion/electrica/costos-de-falla/>
- [22] Pluz Energía Perú S.A.A. "Compensación por norma técnica de calidad de los servicios eléctricos (ntcse)." Comunicado o sección de ayuda de la empresa., Accessed: Oct. 20, 2025. [Online]. Available: <https://www.pluz.pe/es/empresas/compensacion-por-norma-tecnica-de-calidad-de-los-servicios-electricos-ntcse.html>
- [23] Organismo Supervisor de la Inversión en Energía y Minería (OSINERGMIN). "Calidad de suministro." Sección de la Norma Técnica de Calidad de los Servicios Eléctricos (NTCSE)., Gobierno del Perú, Accessed: Oct. 23, 2025. [Online]. Available: <https://www.osinergmin.gob.pe/empresas/electricidad/calidad/NTCSE/suministro>
- [24] *Informe anual: Gestión del operador nacional de electricidad - cenace, 2024*, Informe Anual, Operador Nacional de Electricidad - CENACE, Quito, Ecuador, 2024. [Online]. Available: https://www.cenace.gob.ec/wp-content/uploads/downloads/2025/04/Informe-Anual-CENACE-2024-vf-1-88_c.pdf
- [25] Pacific Gas and Electric Company (PG&E). "Programas de compensación de interrupciones del suministro eléctrico," Accessed: Oct. 23, 2025. [Online]. Available: <https://www.pge.com/es/outages-and-safety/outage-preparedness-and-support/general-outage-resources/outage-compensation-programs.html>
- [26] CNN en Español. "Facturas de electricidad en texas de us\$ 29 millones." Noticia sobre la crisis energética en Texas tras la tormenta de invierno., Accessed: Oct. 23, 2025. [Online]. Available: <https://cnnespanol.cnn.com/2021/03/17/facturas-electricidad-texas-us-29-millones-trax>
- [27] Ofgem (Office of Gas and Electricity Markets). "Ofgem secures customer compensation from ten suppliers for technical overcharging error." Nota de Prensa, Accessed: Oct. 23, 2025. [Online]. Available: <https://ofgem-newsroom.prgloo.com/news/ofgem-secures-customer-compensation-from-ten-suppliers-for-technical-overcharging-error>
- [28] M. Sweney. "Ten uk energy firms to pay £7m in compensation after overcharging error," Accessed: Oct. 23, 2025. [Online]. Available: <https://www.theguardian.com/business/2025/may/09/uk-energy-firms-overcharging-customers-compensation-ofgem>
- [29] S. Estrella Oporto. "Comienza proceso de compensación para clientes afectados por cortes de luz de 2024," Accessed: Oct. 18, 2025. [Online]. Available: <https://www.latercera.com/servicios/noticia/comienza-proceso-de-compensacion-para-clientes-afectados-por-cortes-de-luz-de-2024/>
- [30] P. San Juan. "Enel inicia proceso de compensación a clientes por corte de electricidad de agosto de 2024," Accessed: Oct. 18, 2025. [Online]. Available: <https://www.latercera.com/pulso/noticia/enel-inicia-proceso-de-compensacion-a-clientes-por-corte-de-electricidad-de-agosto-de-2024/>
- [31] Contraloría General de la República. "Revelan perjuicio económico en empresa de electricidad por incumplimientos en compensaciones." Nota de prensa N° 551-2024-CG/CG-GCOC, Accessed: Oct. 23, 2025. [Online]. Available: <https://www.gob.pe/institucion/contraloria/noticias/951548-revelan-perjuicio-economico-en-empresa-de-electricidad-por-incumplimientos-en-compensaciones>
- [32] Diario Crónica. "Eerssa aplica compensación por racionamientos del servicio de energía," Accessed: Oct. 18, 2025. [Online]. Available: <https://cronica.com.ec/2024/12/14/eerssa-aplica-compensacion-por-racionamientos-del-servicio-de-energia/>
- [33] El Comercio. "Compensación en planillas de luz continúa en enero de 2025: ¿cuánto le cuesta al estado?" Accessed: Oct. 18, 2025. [Online]. Available: <https://www.elcomercio.com/actualidad/negocios/compensacion-planillas-luz-continua-enero-2025-cuanto-cuesta-estado/>

VIII. BIOGRAPHY

Leonardo Gabriel Granda Erazo is an Electrical Engineering student at Universidad Nacional de Loja and is currently in the ninth semester of the program. He completed his secondary education at Unidad Educativa Fiscomisional Daniel Álvarez Burneo. His academic interests include electric power systems, stability, programming, and automation.

Contact: leonardo.granda@unl.edu.ec